LEAF DISEASE DETECTION USING MACHINE LEARNING

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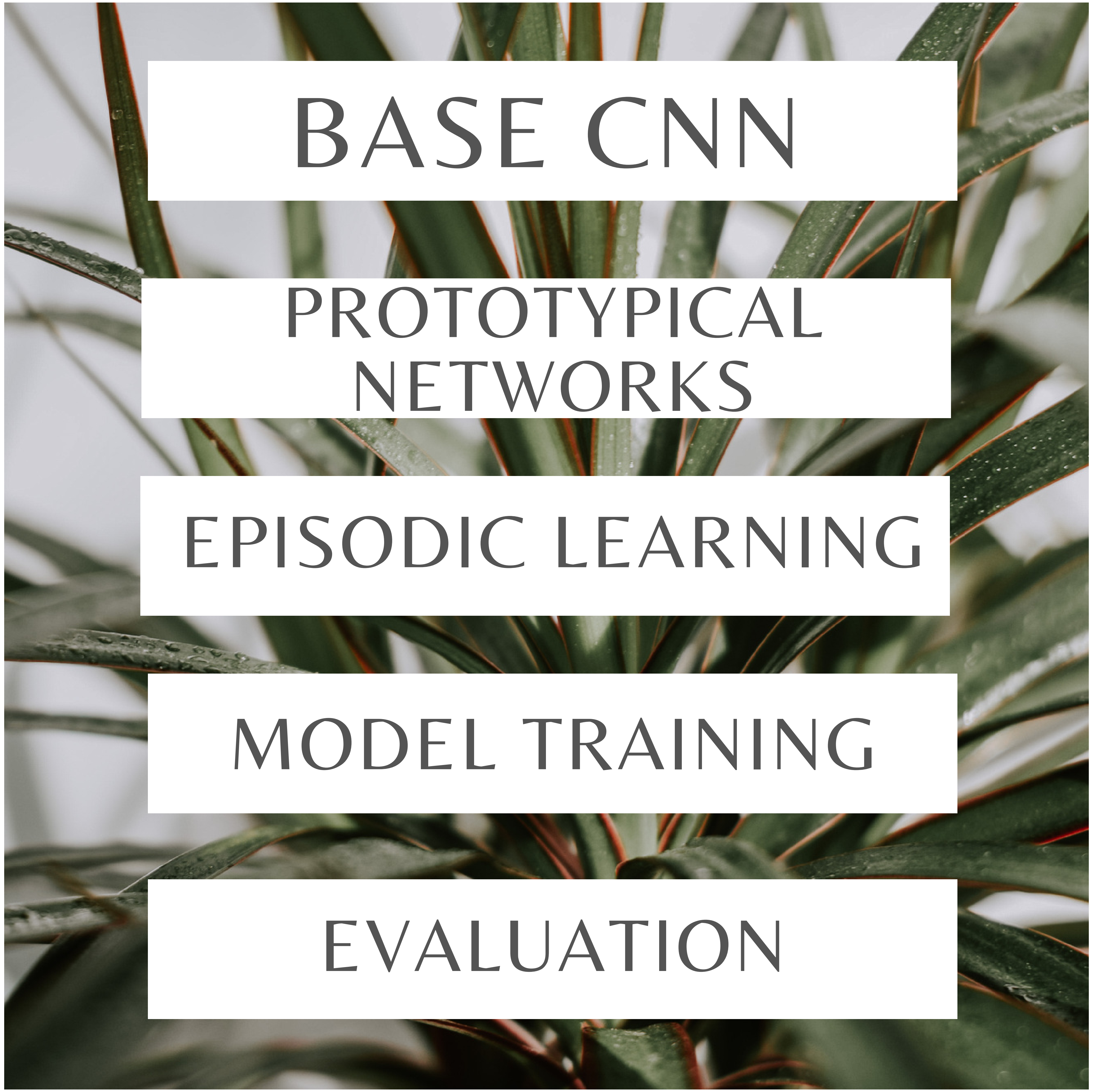
THE WHY?

Plant diseases are a serious concern in Bangladesh, where agriculture supports **40%** of the working population and generates **12.5%** of the **country's GDP**. Each year, they can cause up to **40%** of worldwide agricultural output losses, with an estimated cost of more than **$220 billion**. The growing middle class's need for high-quality farm goods contributes to the effects on smallholder farmers, resulting in losses in income and difficulties in providing food. Our study, "**Leaf Disease Detection using Machine Learning**" intends to solve this issue by offering a practical means of identifying diseases early on. By utilizing machine learning, we want to improve the food security of the world and drastically lower agricultural losses, especially in Bangladesh.

THE HOW?

Our study uses innovative methods of machine learning to identify and categorize plant leaf diseases. The project makes use of **prototypical networks** and **few-shot learning (FSL)**, which work particularly well when there are only a few training examples of a given disease.

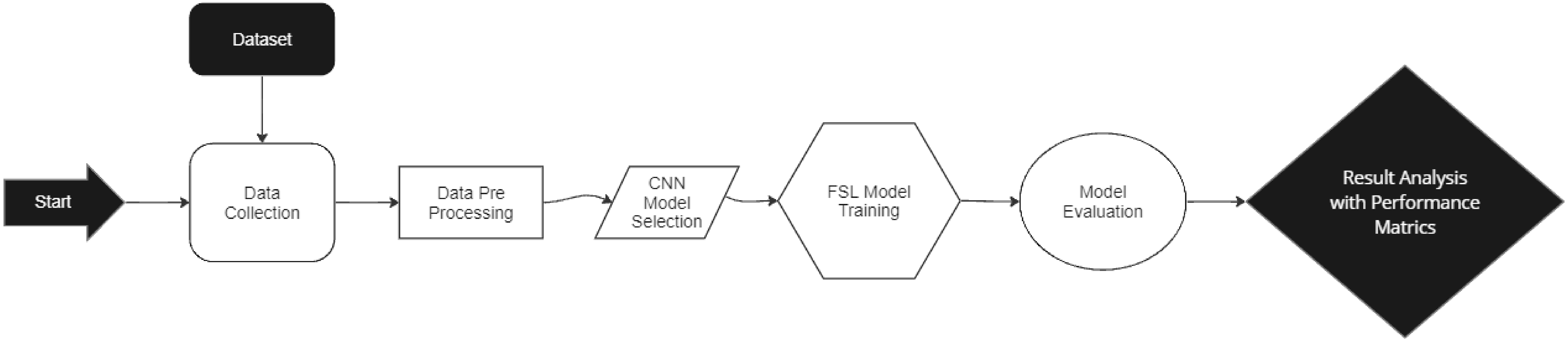
We used a Pre-Augmented **PlantVillage** dataset. We then Resized all the images to **64 x 64 pixels**. Then we changed the file structure for few-shot learning.



THE HOW?

- We use ResNet18 & EfficientNet as Base CNN pretrained model.
- We use a prototypical network to extract image features from support and query set using ResNet18 or EfficientNet CNN.
- To train our few-shot model instead of classical training we use episodic training.
- We create 10,000 or 5,000 few-shot training tasks to train the model on the Training set (image_background). We use Cross Entropy Loss function. We use ADAM optimizer which helps with bias correction & efficiency.
- We create 1,000 evaluation task to evaluate the model on the Evaluation set (images evaluation).

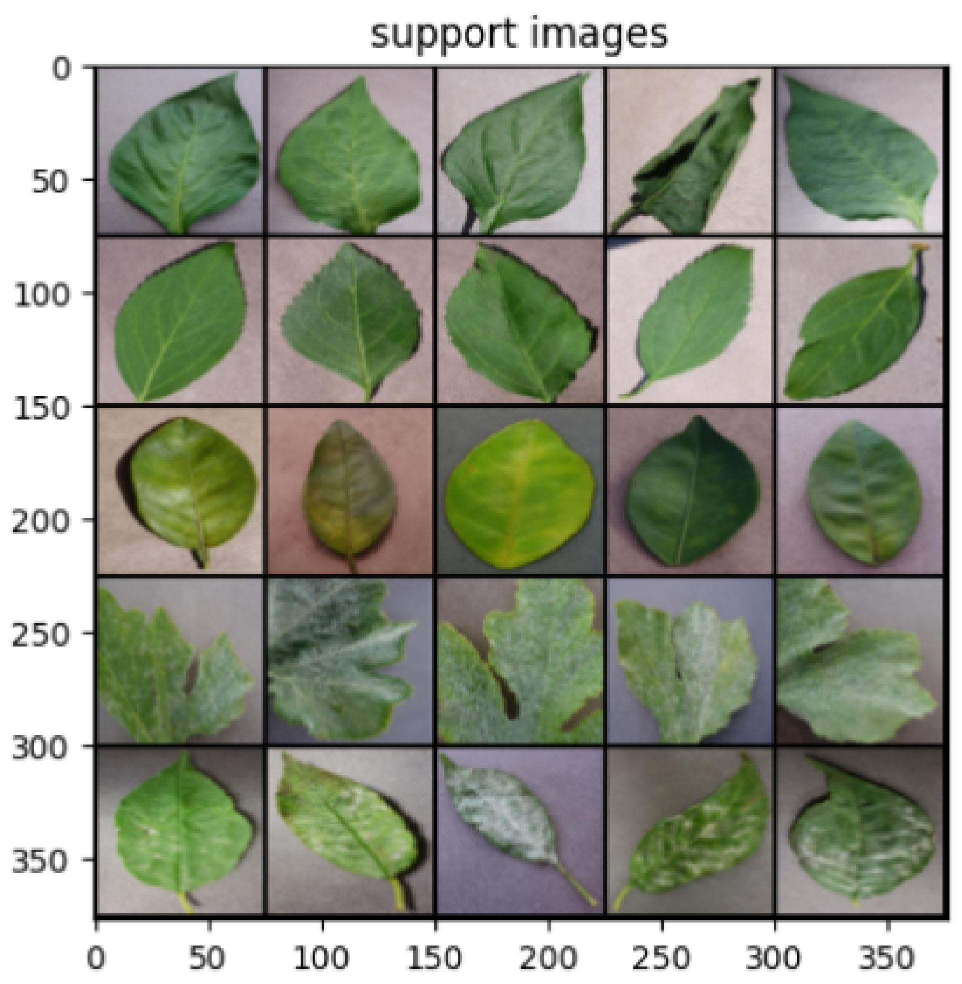
FLOW CHART


Dataset
Start
Data Collection
Data Pre Processing
CNN Model Selection
FSL Model Training
Model Evaluation
Result Analysis with Performance Matrics

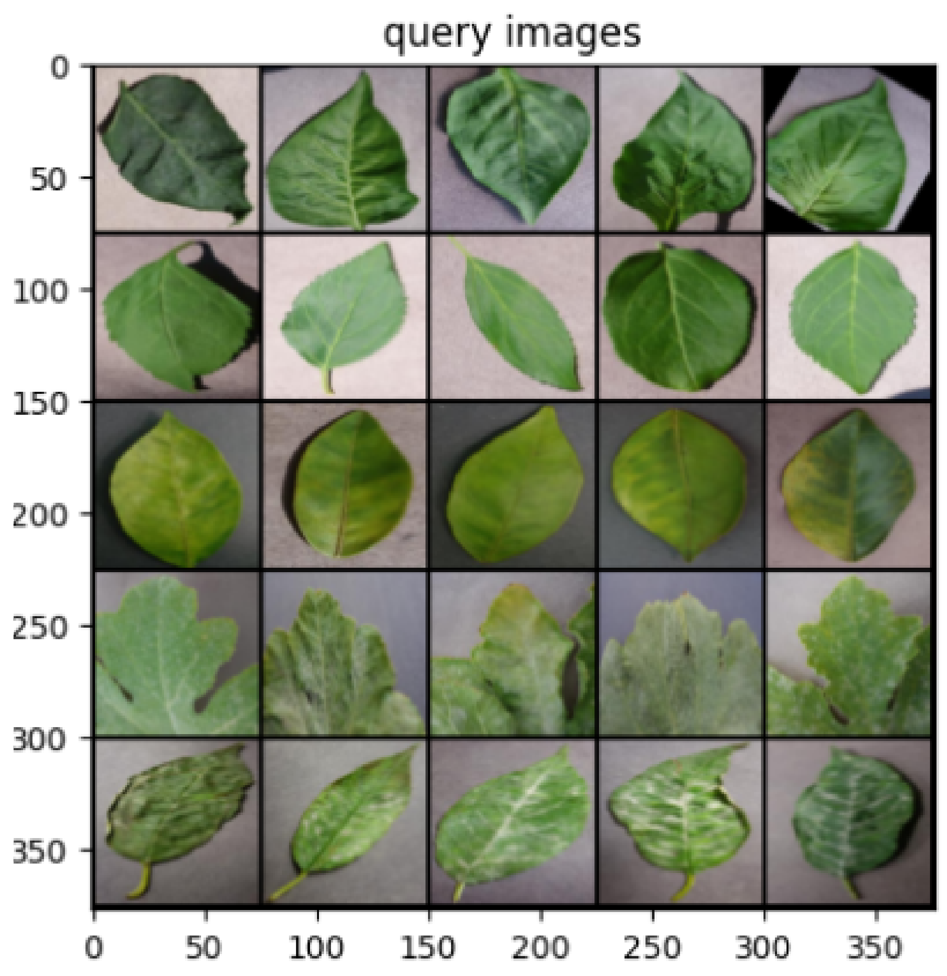
THE HOW?

```
|-- Dataset_Name
|   |-- images_background
|   |   |-- plant_1
|   |   |   |-- leaf_1
|   |   |   |   |-- image_1.JPG
|   |   |   |   |-- image_2.JPG
|   |   |   |   |-- ...
|   |   |   |-- leaf_2
|   |   |   |   |-- image_1.JPG
|   |   |   |   |-- image_2.JPG
|   |   |   |   |-- ...
|   |   |   |-- ...
|   |   |-- plant_2
|   |   |   |-- ...
|   |   |-- ...
|   |-- images_evaluation
|   |   |-- plant_1
|   |   |   |-- ...
|   |   |-- plant_2
|   |   |   |-- ...
|   |   |-- ...
```

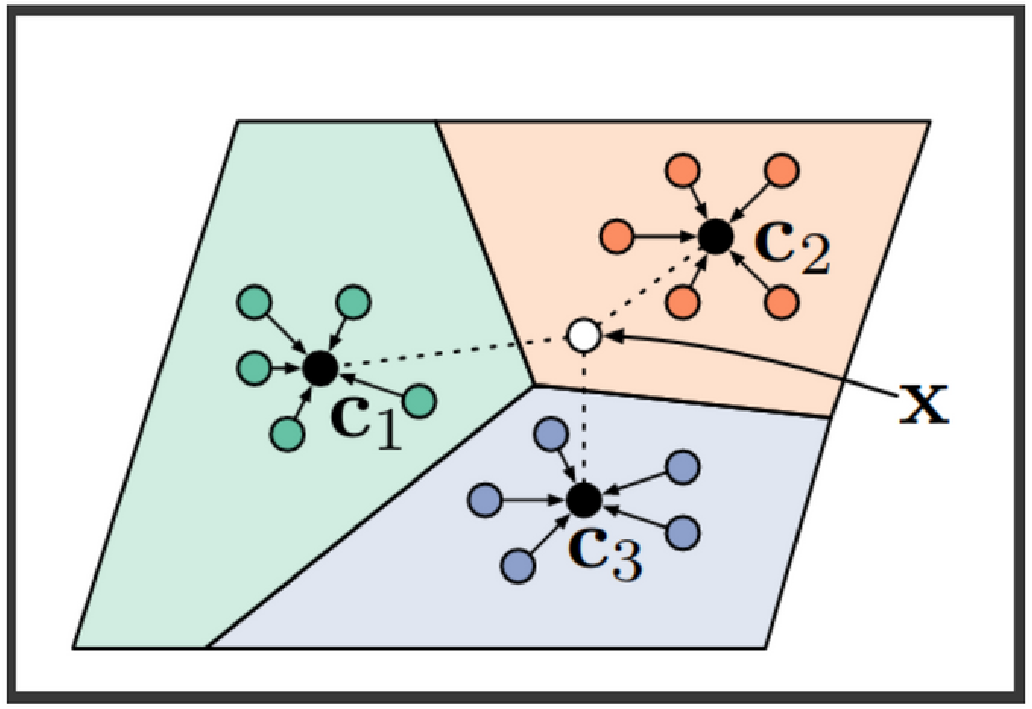
- We first use a Custom dataset class to create a dataset object from any dataset with a file tree.
- Then we divide the dataset to train and test set. Where training set = images_background & test set = images_evaluation.
- Then we extract the image embeddings using either EfficientNet or Resnet18 pre-trained CNN and put images of the same class in the same feature space.
- After that we use prototypical network to create prototypes for each class from the support set. These prototypes are the mean of all embeddings of support images from the same class.
- Then we create our train and test loader using “DataLoader” method from easyFSL to feed few-shot classification tasks to the model.
- As we use episodic training, we train our model over a large number of randomly generated few-shot classification tasks using CrossEntropyLoss as a Loss function and ADAM as the optimizer.
- After that we evaluate our few-shot model based on Accuracy, Top-k Accuracy, Precision, Recall, F1 score & AUC.



Support Set



Query Set



Feature Extraction Using Prototypical Network

TABLE I
MODEL ACCURACY & AUC

CNN	Training Episode	Accuracy in %	AUC
EfficientNet	10,000	77.91	0.954
ResNet18	10,000	68.25	0.913
EfficientNet	5,000	81.32	0.964
ResNet18	5,000	73.55	0.933

TABLE II
MODEL PRECISION, RECALL & F1 SCORE

CNN	Training Episode	Precision	Recall	F1 Score
EfficientNet	10,000	0.71	0.74	0.73
ResNet18	10,000	0.73	0.67	0.70
EfficientNet	5,000	0.83	0.74	0.78
ResNet18	5,000	0.70	0.64	0.67

RESULT



THANK YOU